

<https://www.ti.com/lit/ds/symlink/opa818.pdf>
p27

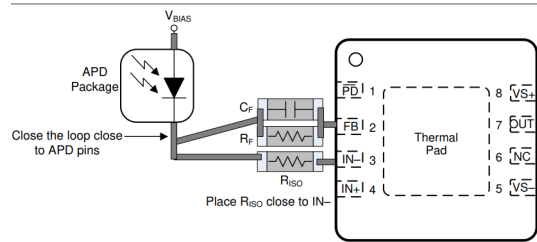


Figure 60. Improved TIA Layout

Photodiodes with TO-18 footprint:

Farnell 1495575
Hamamatsu S5973
D= 400um active area
Cd=1.6 pF @ Vr=3V3

Thorlabs FDS015
D=150 um active area
0.65pF @ Vr = 5 V

Thorlabs FDS025
D= 250 um active area, with ball lens
0.94 pF @ Vr = 5 V

Hamamatsu S5972
D= 800um active area
3 pF @ Vr = 10 V

Hamamatsu S5971
D= 1.2 mm active area
3 pF @ Vr = 10 V

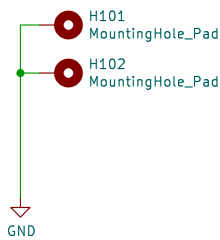
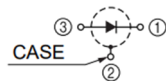
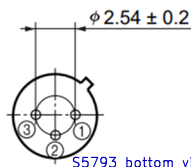
OSI FCI-125G-006HRL
Hamamatsu S9055(-01)

Thorlabs FGA01FC, InGaAs, 2pF @ 5V
same as OSI INGAAS-120L-FC

Thorlabs FGA015, InGaAs, 1.5 pF @ 5V
D = 0.15 mm active area

FGA01, ball lens

For estimating Bandwidth, Transimpedance-amplifier noise, shot-noise limit, etc. see:
TIA_Sim at <https://github.com/aewallin/TIASim>



PSU:
Thorlabs LDS12B
±12 VDC Regulated Linear Power Supply, 6 W
Lumberg RSMV3 male connector, 3-pin

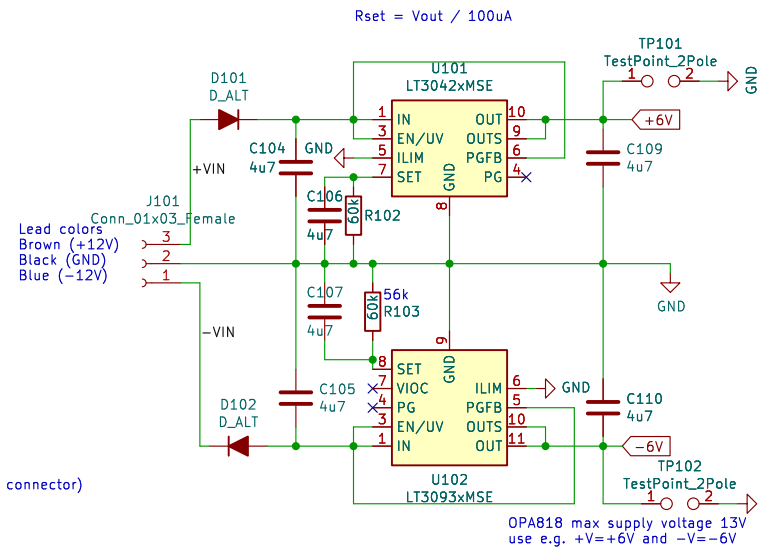
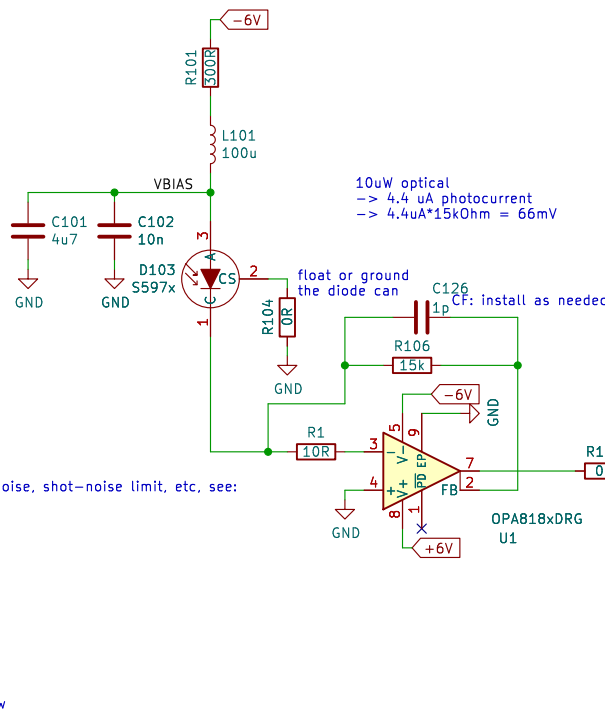
PSU-lead: Digikiy A120947-ND (mates with Thorlabs 3-pin psu connector)
3-pin M8 female connector
1.5 m 3-wire lead, solder to PCB

See MeasurementData sheet for measurements and
noise-models for built 2-stage detectors:

MeasurementData



File: 2stageTIA_measurementData.kicad_sch



1	2	3	4	5	6
A					A
B					B
C					C
D					D
1	2	3	4	5	6

Sheet: /MeasurementData/ File: 2stageTIA_measurementData.kicad_sch		
Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. eeschema 7.0.6	Id: 2/2	